

Econ 101A

Section 7 (Solutions)

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Now that we can solve the consumer problem, this section asks how the answer *moves*. Two questions. How does demand respond to income and to prices (comparative statics)? And what is the other, “dual” way to describe the same consumer through cost rather than utility (expenditure minimization)? These are the core tools of demand theory, and the exercises walk the steps on clean, tractable functions.

1 Comparative Statics of Demand

Marshallian demand $x^*(p, q, M)$ is homogeneous of degree zero: scaling all prices and income by the same factor changes nothing. Two slices matter.

Income (the Engel curve). Fix prices and vary M . The good is

$$\text{normal if } \frac{\partial x^*}{\partial M} > 0, \quad \text{inferior if } \frac{\partial x^*}{\partial M} < 0.$$

Own price (the demand curve). Fix M and the other price. Almost always $\partial x^*/\partial p < 0$. The exception is a **Giffen** good, where $\partial x^*/\partial p > 0$. A cross-price effect $\partial x^*/\partial q$ classifies x and y as **gross substitutes** (> 0) or **gross complements** (< 0).

2 The Dual Problem: Expenditure Minimization

Instead of maximizing utility for a given budget, fix a target utility $\bar{u}$ and spend as little as possible to reach it:

$$\min_{x,y} px + qy \quad \text{s.t.} \quad u(x, y) \geq \bar{u}.$$

The solution is **Hicksian (compensated) demand** $h_x(p, q, \bar{u})$, $h_y(p, q, \bar{u})$, and the minimized cost is the **expenditure function**

$$e(p, q, \bar{u}) = p h_x + q h_y.$$

The tangency condition is the same as in utility maximization ($u_x/u_y = p/q$); only the constraint differs, so the multiplier differs.

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Duality. The two problems describe the same consumer:

$$h_i(p, q, \bar{u}) = x_i^*(p, q, e(p, q, \bar{u})), \quad x_i^*(p, q, M) = h_i(p, q, v(p, q, M)).$$

Two facts you will use constantly.

- **Compensated law of demand:** $\partial h_i / \partial p_i \leq 0$, always. Holding utility fixed, a higher own price cannot raise compensated demand. No Giffen paradox on the Hicksian curve.
- **Shephard's lemma (envelope):** applying the envelope theorem to the EMIN Lagrangian $\mathcal{L}_E = px + qy + \mu(\bar{u} - u)$ gives

$$\frac{\partial e}{\partial p} = \left. \frac{\partial \mathcal{L}_E}{\partial p} \right|_{\text{opt}} = h_x.$$

The extra cost of a small price rise is just the quantity you were buying; the reoptimization terms cancel at the optimum.

3 The Slutsky Equation: Substitution and Income

The bridge between the two demands. For a change in the own price,

$$\underbrace{\frac{\partial x^*}{\partial p}}_{\text{total (Marshallian)}} = \underbrace{\left. \frac{\partial h_x}{\partial p} \right|_{\bar{u}=v}}_{\text{substitution} \leq 0} - \underbrace{x^* \frac{\partial x^*}{\partial M}}_{\text{income}}.$$

The substitution term is never positive. So the only way to get a Giffen good ($\partial x^* / \partial p > 0$) is a negative, dominant income term, which needs $\partial x^* / \partial M < 0$: **every Giffen good is inferior**. The converse fails: a good can be inferior with a small income effect that never overturns the substitution effect.

4 Exercises

Exercise 1 (expenditure minimization and duality)

Let $u(x, y) = \sqrt{xy}$ with prices $p, q > 0$. (These are the same preferences as Section 6, where $x^* = M/2p$, $y^* = M/2q$, $v = M/(2\sqrt{pq})$.)

- Solve the expenditure minimization problem for the Hicksian demands $h_x(p, q, \bar{u})$ and $h_y(p, q, \bar{u})$.
- Compute the expenditure function $e(p, q, \bar{u})$.
- Verify duality: show $h_x = x^*(p, q, e)$.
- Verify Shephard's lemma: check $\partial e / \partial p = h_x$.

Solution. (a) Minimize $px + qy$ s.t. $\sqrt{xy} \geq \bar{u}$; the constraint binds. Tangency (same as in Section 6) gives $qy = px$, i.e. $y = \frac{p}{q}x$. Impose $\sqrt{xy} = \bar{u}$: $\sqrt{x \cdot (p/q)x} = x\sqrt{p/q} = \bar{u}$, so

$$h_x = \bar{u}\sqrt{q/p}, \quad h_y = \bar{u}\sqrt{p/q}.$$

(b) $e = ph_x + qh_y = \bar{u}(p\sqrt{q/p} + q\sqrt{p/q}) = \bar{u}(\sqrt{pq} + \sqrt{pq}) = 2\bar{u}\sqrt{pq}$.

(c) Plug $M = e = 2\bar{u}\sqrt{pq}$ into the Marshallian $x^* = M/2p$: $x^* = \frac{2\bar{u}\sqrt{pq}}{2p} = \bar{u}\sqrt{q/p} = h_x$. ✓

(d) $\frac{\partial e}{\partial p} = 2\bar{u} \cdot \frac{\partial}{\partial p}\sqrt{pq} = 2\bar{u} \cdot \frac{\sqrt{q}}{2\sqrt{p}} = \bar{u}\sqrt{q/p} = h_x$. ✓

Exercise 2 (Giffen versus inferior)

Use the Slutsky equation to answer, with derivatives, not words alone.

- (a) Write the Slutsky equation for $\partial x^*/\partial p$ and state the sign of the substitution term and why.
- (b) Show that if x is Giffen (with $x^* > 0$) then x must be inferior.
- (c) Show the converse is false: an inferior good need not be Giffen.

Solution. (a) $\frac{\partial x^*}{\partial p} = \frac{\partial h_x}{\partial p} \Big|_{\bar{u}=v} - x^* \frac{\partial x^*}{\partial M}$. The substitution term $\partial h_x/\partial p \leq 0$ by the compensated law of demand: holding utility fixed, a higher own price shifts the compensated bundle away from x .

(b) Giffen means $\partial x^*/\partial p > 0$. Since the substitution term is ≤ 0 , the income piece $-x^* \partial x^*/\partial M$ must be positive and large enough to dominate. With $x^* > 0$ that requires $\partial x^*/\partial M < 0$: the good is inferior.

(c) Inferior only means $\partial x^*/\partial M < 0$. If $x^* |\partial x^*/\partial M|$ is smaller than $|\partial h_x/\partial p|$, the total stays negative and the good is a normal-looking downward-sloping (ordinary) demand despite being inferior. Inferiority is necessary for Giffen, not sufficient.

Exercise 3 (numerical Slutsky decomposition)

Keep $u(x, y) = \sqrt{xy}$, and evaluate at $p = q = 1$, $M = 8$.

- (a) Find the initial bundle (x^*, y^*) and utility u_0 .
- (b) Compute the substitution term $\partial h_x/\partial p$ (holding $\bar{u} = u_0$) and the income term $-x^* \partial x^*/\partial M$.
- (c) Add them to get $\partial x^*/\partial p$, check against direct differentiation of the Marshallian demand, and say whether x is Giffen.

Solution. (a) $x^* = M/2p = 4$, $y^* = M/2q = 4$, $u_0 = \sqrt{16} = 4$.

(b) Substitution: $h_x = \bar{u}\sqrt{q/p} = \bar{u}q^{1/2}p^{-1/2}$, so $\frac{\partial h_x}{\partial p} = -\frac{\bar{u}\sqrt{q}}{2p^{3/2}}$. At $\bar{u} = 4$, $p = q = 1$: $\partial h_x/\partial p =$

$$-\frac{4}{2} = -2.$$

Income: $x^* = M/2p$ gives $\partial x^*/\partial M = 1/(2p) = 1/2$, so $-x^* \partial x^*/\partial M = -4 \cdot \frac{1}{2} = -2$.

(c) Total $= -2 + (-2) = -4$. Directly, $x^* = M/2p \Rightarrow \partial x^*/\partial p = -M/(2p^2) = -8/2 = -4$. ✓ Both pieces are negative and x is normal ($\partial x^*/\partial M > 0$), so x is an ordinary good, not Giffen.